Problem Set 3 — Generalized linear models

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show derivations and give every number with a sentence of interpretation in context. Use `set.seed(2027)` whenever you simulate or split data. Problem 7 needs the package `pscl` (`install.packages("pscl")`). Notation as in the notes: $f(y;\theta,\phi)=\exp\{[y\theta-b(\theta)]/a(\phi)+c(y,\phi)\}$, $\mu=b'(\theta)$, $V(\mu)=b''(\theta)$, $g(\mu)=\eta=X\beta$, k is the negative binomial size, $\hat\phi=X^2/(n-p)$. Tables: $\chi^2_{1;0.95}=3.84$, $\chi^2_{1;0.99}=6.63$, $F_{1,23;0.99}=7.88$.

1 The zero-truncated Poisson as an exponential family

A count Y that can only be observed when it is positive (for example, the number of items per purchase, among customers who bought something) has the *zero-truncated Poisson* distribution

$$P(Y=y)=\frac{\lambda^y e^{-\lambda}}{y!\,(1-e^{-\lambda})},\qquad y=1,2,\ldots,\ \lambda>0.$$

(a) Write this distribution in the form $f(y;\theta,\phi)=\exp\{[y\theta-b(\theta)]/a(\phi)+c(y,\phi)\}$ and give θ, $b(\theta)$, $a(\phi)$ and $c(y,\phi)$.

(b) Use $b(\theta)$ to show that $\mathrm{E}[Y]=\mu=\lambda/(1-e^{-\lambda})$ and $\mathrm{Var}(Y)=\mu(1+\lambda-\mu)$.

(c) Show that $\mathrm{Var}(Y)<\mathrm{E}[Y]$ for every $\lambda>0$. (Hint: show $\mu-\lambda=\lambda/(e^\lambda-1)<1$.) Evaluate the mean and variance for $\lambda=2$ and check them numerically in R by summing the probability mass function.

(d) Consider the regression model $\log\lambda_i=\boldsymbol{x}_i^\top\beta$ for independent truncated counts. Show that it is the canonical-link GLM for this family (in terms of θ), and write the likelihood equations. Why is $\theta=\log\lambda$ not a function of μ that you can write in closed form?

(e) *Simulation.* Simulate $n=5000$ pairs with $x_i\sim U(0,1)$ and Poisson counts with $\lambda_i=\exp(-0.5+x_i)$, and keep only the positive counts. Fit (i) an ordinary Poisson regression to the kept data and (ii) the truncated model, by maximizing the truncated log-likelihood with `optim`. Compare both fits with the true values $(-0.5,\ 1)$ and explain.

2 Fisher scoring with the identity link

A Poisson regression with the **identity link**, $\mu_i=\beta_0+\beta_1 x_i$, is used when the effect of x is thought to be additive on the mean. Data: $x=(1,2,3,4,5)$, $y=(2,3,6,7,10)$.

(a) Give $g'(\mu)$, the IRLS weight w_i and the working response z_i for this model. Show that $z_i=y_i$ for every iteration, so that each IRLS step is a weighted least-squares fit of y on x with weights $1/\hat\mu_i$.

(b) Start from $\hat\mu_i^{(0)}=y_i$. Compute by hand the weighted sums $\sum w_i$, $\sum w_i x_i$, $\sum w_i x_i^2$, $\sum w_i y_i$, $\sum w_i x_i y_i$ and the estimate $\hat\beta^{(1)}$.

(c) Run `glm(y ~ x, family = poisson(link = "identity"))` and compare with (b). How many Fisher scoring iterations did R need?

(d) Derive the observed information of this model, $I(\beta)=\sum_i (y_i/\mu_i^2)\,\boldsymbol{x}_i\boldsymbol{x}_i^\top$, from the general formula of Chapter 9, and compare numerically the standard errors from the observed and from the expected information at $\hat\beta$. Is IRLS here Newton–Raphson?

(e) Show from the likelihood equations that $\sum_i (y_i-\hat\mu_i)=0$ although the link is not the canonical one.

(f) What goes wrong if you predict at $x=-2$? Which link would you use instead?

3 Likelihood-ratio and dispersion calculations from printed output (chromosome aberrations)

The data `faraway::dicentric` give, for 27 irradiation experiments, the number of cells examined (`cells`), the number of cells with chromosome aberrations (`ca`), the dose amount in Gy (`doseamt`: 1, 2.5, 5) and the dose rate in Gy/hour (`doserate`, nine values from 0.1 to 4). Two Poisson models with offset log(`cells`) were fitted. The relevant output is printed below; **answer (a)–(g) from the printed output** (a calculator and the tables above are enough).

```
#> Model A: ca ~ offset(log(cells)) + log(doseamt) + log(doserate)
#>   Residual deviance: 44.86 on 24 df   AIC: 226.3   Pearson X2: 44.04
#>
#> Model B: ca ~ offset(log(cells)) + log(doseamt) * log(doserate)
#>   Residual deviance: 26.11 on 23 df   AIC: 209.5   Pearson X2: 25.96
#>                             Estimate Std. Error  z value Pr(>|z|)
#> (Intercept)                  -2.7292     0.0318 -85.7034   0.0000
#> log(doseamt)                  1.7115     0.0268  63.8803   0.0000
#> log(doserate)                 0.0925     0.0323   2.8683   0.0041
#> log(doseamt):log(doserate)    0.1130     0.0260   4.3440   0.0000
```

(a) Write the model B for $\log \mathrm{E}[\mathtt{ca}_i]$ and say what the coefficient of the interaction term means.

(b) Test the interaction with a likelihood-ratio test (statistic, degrees of freedom, conclusion at 5%). Check that the AIC difference agrees with the deviance difference.

(c) Estimate the dispersion of model B by the Pearson method and by the deviance method. Do an F-test of the interaction using the Pearson estimate and compare with (b).

(d) Give the rate ratio for *doubling the dose amount* at dose rate 1 Gy/hour and at dose rate 4 Gy/hour, and the rate ratio for *doubling the dose rate* at dose amount 1 Gy and 5 Gy. (Remember that the covariates enter as logarithms.)

(e) Compute a 95% Wald interval for the interaction coefficient. Is it compatible with the conclusion of (b)?

(f) Predict the expected number of aberrations among 1000 cells irradiated with 2.5 Gy at 1 Gy/hour. Compute the Pearson residual of experiment 2 (1907 cells, 102 aberrations, 1 Gy, 0.25 Gy/hour).

(g) Model A has Pearson $X^2 = 44.04$ on 24 df. What does this say about overdispersion in model A and in model B, and what does it suggest about the interaction?

4 Logistic regression and classification without leakage (breast biopsies)

`MASS::biopsy` contains nine cytological scores (`V1`–`V9`, each from 1 to 10) for 699 breast tumours and the diagnosis `class` (benign or malignant). We use the 683 complete cases, code `y = 1` for malignant, and split them at random into 70% training and 30% test data.

```
bio <- biopsy[complete.cases(biopsy), -1]
bio$y <- as.integer(bio$class == "malignant"); bio$class <- NULL
set.seed(2027); idx <- sample(nrow(bio), round(0.7 * nrow(bio)))
train <- bio[idx, ]; test <- bio[-idx, ]
c(train = nrow(train), test = nrow(test), prevalence_train = mean(train$y),
  prevalence_test = mean(test$y))
```

```
#>           train            test prevalence_train  prevalence_test
#>       478.00000       205.00000         0.34937         0.35122
```

(a) Fit the logistic regression of `y` on `V1`–`V9` using the **training** data. Report the odds ratios with 95% profile intervals for `V1` and `V6` and interpret them in one sentence each. Which scores are not significant?

(b) Test with a likelihood-ratio test whether the block `V2 + V3 + V5` (3 df) can be dropped. State the hypotheses, statistic, degrees of freedom and conclusion.

(c) A colleague reports that "the residual deviance is far below its degrees of freedom, so the model fits well". Explain why this statement is meaningless for binary data, and instead produce a calibration table on the **test** data (five groups of fitted probabilities: mean predicted and observed frequency).

(d) Compute the test AUC with a function you write yourself from the rank (Mann–Whitney) formula, and compare it with the training AUC.

(e) A false negative (a malignant tumour called benign) is judged ten times as costly as a false positive. Give the cost-optimal threshold c^*, and compute the confusion matrix and total cost $C_{FP}\cdot\mathrm{FP}+C_{FN}\cdot\mathrm{FN}$ (with $C_{FP}=1$) on the test data for $c=0.5$ and $c=c^*$.

(f) Suppose instead that no costs are available and the team proposes Youden's index. Choose the cut-off (i) on the training data, (ii) by 5-fold cross-validation on the training data (`set.seed(2027)`), (iii) on the test data. Report the test sensitivity and specificity for each and explain which ones may be quoted as the performance of the classifier.

(g) Write a short paragraph (at most 6 sentences) for the clinical team: what can they claim, what can they not, and what must they check before using the model on patients from another hospital?

5 Prevalence shift and the threshold

The classifier of Problem 4 was trained on biopsies in which about 35% of tumours are malignant. A screening programme will apply it to women among whom only 5% of the tested lesions are malignant.

(a) Let π_s be the prevalence in the training data and π_p in the target population, and suppose that the distribution of the scores *given the diagnosis* is the same in both. Show that the posterior odds in the population are

$$\frac{P_p(Y=1\mid \boldsymbol{x})}{P_p(Y=0\mid \boldsymbol{x})} = \frac{P_s(Y=1\mid \boldsymbol{x})}{P_s(Y=0\mid \boldsymbol{x})} \times \frac{\pi_p/(1-\pi_p)}{\pi_s/(1-\pi_s)}.$$

(b) A woman receives a fitted probability $\hat{\mu}=0.30$ from the model trained with prevalence $\pi_s=0.35$. What is her probability of malignancy in the screening population? Does this change the decision for $C_{FN}=10\,C_{FP}$?

(c) Use sensitivity 0.97 and specificity 0.95 at some threshold (from Problem 4). Compute the positive predictive value in the training population ($\pi_s=0.35$) and in the screening population ($\pi_p=0.05$), per 1000 women screened in each case: numbers of true positives, false positives, and PPV.

(d) *R.* Apply the correction of (a) to the test-set probabilities of Problem 4 (use the training prevalence for π_s and $\pi_p=0.05$). With $C_{FN}=10\,C_{FP}$, how many of the 205 test patients would be flagged using the corrected probabilities and $c^*=1/11$, compared with the uncorrected ones?

(e) Explain in two sentences why the cost-optimal threshold does not depend on the prevalence but the decision of (d) does.

6 Overdispersion and clustering: epileptic seizures

`MASS::epil` records the number of epileptic seizures `y` of 59 patients in each of four consecutive two-week periods, with the 8-week baseline count `base`, the treatment (`trt`: placebo or progabide) and `age`. We model y with the covariates log(`base`/4) (baseline seizure rate per two weeks), `trt`, log(`age`) and an indicator `V4` for the fourth period.

```
epil$lbase <- log(epil$base / 4); epil$lage <- log(epil$age); epil$V4 <-
↪  as.integer(epil$period == 4)
c(n = nrow(epil), patients = length(unique(epil$subject)), mean = mean(epil$y),
↪  variance = var(epil$y))
```

```
#>        n patients     mean variance
#> 236.0000  59.0000   8.2627 152.6796
```

(a) Fit the Poisson regression `y ~ lbase + trt + lage + V4`. Report $\hat{\phi} = X^2/(n-p)$ and carry out the one-sided Cameron–Trivedi test ($\mathrm{Var}(Y) = \mu + \alpha\mu^2$). Conclusion?

(b) Fit quasi-Poisson and negative binomial models with the same linear predictor. Tabulate the coefficient, standard error and p-value of `trt` in the three models. Explain why the standard error is the same multiple of the Poisson one for all coefficients in the quasi-Poisson model, and why the estimates of the NB model differ from the Poisson ones.

(c) The two remedies give different conclusions about the treatment. Plot the cell variance against the cell mean (for example, grouping observations by deciles of the fitted mean) together with the quasi-Poisson and NB variance curves. Which variance function is supported?

(d) The four observations of a patient are not independent. Compute the 4×4 correlation matrix of the Pearson residuals of the Poisson fit across the four periods (use `matrix(residuals(fit, "pearson"), ncol = 4, byrow = TRUE)`). What does it show, and why does it matter for the standard errors of `trt`, which is a patient-level covariate?

(e) Compute randomized quantile residuals (`set.seed(2027)`) for the Poisson and the NB models and report the range and a QQ-plot for each.

(f) Which of the three analyses of the treatment effect would you trust, and what would you do instead (name the class of models of Chapter 13, without fitting it)? Write a summary of at most 150 words of the treatment effect and its uncertainty for a clinician who does not know statistics.

7 Excess zeros: scientific articles of PhD students

`pscl::bioChemists` records the number of articles `art` published during the last three years of their PhD by 915 biochemistry students, with gender (`fem`), marital status (`mar`), number of children under 6 (`kid5`), prestige of the PhD department (`phd`) and the number of articles of the mentor (`ment`).

```
library(pscl)
c(n = nrow(bioChemists), zeros = sum(bioChemists$art == 0), mean =
↪  mean(bioChemists$art), variance = var(bioChemists$art))
```

```
#>        n    zeros     mean variance
#> 915.0000 275.0000   1.6929   3.7097
```

(a) Fit the Poisson regression `art ~ fem + mar + kid5 + phd + ment`. Compute $\hat{\phi}$ and the expected number of zeros under the fitted model. Compare with the observed number. Why is `dpois(0, mean(art))` not the right comparison?

(b) Fit the negative binomial model, the zero-inflated Poisson (ZIP), the zero-inflated negative binomial (ZINB) and the hurdle Poisson model (`zeroinfl`, `hurdle`, same covariates in both parts). Tabulate AIC and the expected number of zeros of all five models.

(c) State the null and alternative hypotheses for testing Poisson against NB, explain why the null value lies on the boundary of the parameter space, and compute the LR statistic and the p-value from the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$.

(d) Show that for the ZIP model $\mathrm{E}[Y] = (1-\pi)\mu$ and $\mathrm{Var}(Y) = (1-\pi)\mu(1+\pi\mu)$. Compute them, and $P(Y=0)$, for $\pi = 0.3$ and $\mu = 2$.

(e) Show that the log-likelihood of the hurdle model separates into a binary part and a zero-truncated count part, so that the two can be maximized separately.

(f) Is there evidence of zero inflation in these data? Which model would you report for the effect of `ment` and of `kid5`, as rate ratios with 95% intervals, and why?

8 Spot the errors

The following is an excerpt of a draft report. Each of the numbered statements contains a conceptual or numerical error. For each, (i) say what is wrong, (ii) give the correct statement or the correct number, and (iii) say how the conclusion would change, if at all.

1. "The logistic model for low birth weight has residual deviance 196.8 on 180 degrees of freedom; as the deviance is close to its degrees of freedom the model fits well."
2. "The odds ratio for smoking is 2.33, so smokers are 2.3 times as likely to have a low-birth-weight baby."
3. "We chose the cut-off 0.22 that maximizes Youden's index on the test set; the resulting test sensitivity (0.89) and specificity (0.68) are our estimates of the classifier's performance."
4. "The Poisson model for ship incidents has residual deviance 38.70 on 25 df. We estimated the dispersion by $\hat{\phi} = 38.70/25 = 1.55$, as quasi-Poisson does, so quasi-Poisson standard errors are $\sqrt{1.55} = 1.24$ times the Poisson ones." (The Pearson statistic is $X^2 = 42.28$.)
5. "Because we used the probit link for the binary response, `glm()` fitted the model by Newton–Raphson."
6. "The Vuong test comparing Poisson with ZIP gave $p < 0.001$, so the Poisson model is rejected in favour of zero inflation."
7. "The negative binomial fit reports `theta` = 1.27, which is the dispersion parameter: a value close to 1 indicates little overdispersion."
8. "A coefficient of 0.40 in the logistic model means that the probability of the event increases by 40 percentage points per unit of the covariate."

9 Randomized quantile residuals

Let Y be a continuous response with a continuous, strictly increasing distribution function F, and let $\hat{F}_i$ be the fitted distribution function of observation i.

(a) Show that if the model is correct, $F(Y) \sim U(0,1)$ and hence $\Phi^{-1}\{F(Y)\} \sim N(0,1)$.

(b) A count has fitted distribution Poisson($\hat{\mu} = 3$) and is observed as $y = 5$. Give the interval in which the randomized quantile residual lies ($P(Y \le 4) = 0.8153$, $P(Y \le 5) = 0.9161$). Repeat for $y = 0$.

(c) *Simulation.* Generate $n = 300$ counts from the model $\log \mu_i = 0.5 + x_i$, $x_i \sim U(0,1)$, $Y_i \sim \text{Poisson}(\mu_i)$ (`set.seed(2027)`), fit the Poisson regression and compute Pearson, deviance and randomized quantile residuals. Report the Shapiro–Wilk p-value of each. Repeat with negative binomial counts of size $k = 2$ (same means) and the *Poisson* fit. Which residuals detect the overdispersion most clearly?